

Data Center and Virtualization

1.	Data Center Concepts & Architecture ✓ Data Center Definition, Purpose, Types of Data Centers, Data Center Architecture, Physical vs Virtual Data Center, Data Center Infrastructure, Data Center Network Architecture, Server Infrastructure, Storage Infrastructure, Power Infrastructure, Cooling Infrastructure, Physical Security, Environmental Monitoring.	1 Day
2.	✓ Data Center Tiers ✓ Tier I, Tier II, Tier III, Tier IV, Availability, Redundancy, Fault Tolerance, Maintenance Capability, Downtime Comparison, Tier Comparison.	1 Day
3.	✓ RAID Levels ✓ RAID Concept, Disk Striping, Disk Mirroring, Parity, Fault Tolerance, RAID 0, RAID 1, RAID 5, RAID 6, RAID 10, Usable Capacity, Minimum Number of Disks, Performance, Fault Tolerance, Advantages, Disadvantages, RAID Comparison.	1.5 Day
4.	✓ Data Center Components ✓ Rack: Rack Structure, Rack Units (U), Rack Mounting, Rack Layout. ✓ Server: Rack Server, Blade Server, Tower Server, Server Components. ✓ Storage: HDD, SSD, Storage Arrays, Storage Capacity. ✓ Cooling: HVAC, CRAC, CRAH, Airflow Management, and Temperature & Humidity Control. ✓ Power: UPS, PDU, Power Distribution, Generator, Power Backup, Power Monitoring.	0.5 Day
5.	✓ Redundancy ✓ Redundancy Concept, N Configuration, N+1 Configuration, 2N Configuration, 2N+1 Configuration, Power Redundancy, Cooling Redundancy, Component Redundancy, Single Point of Failure, Redundancy Comparison.	0.5 Day
6.	✓ Hot Aisle vs Cold Aisle ✓ Hot Aisle, Cold Aisle, Rack Arrangement, Airflow Direction, Supply Air, Return Air, Containment, Hot Aisle Containment, Cold Aisle Containment, Cooling Efficiency, Energy Savings, Comparison.	0.5 Day

7.	✓ Storage Technologies ✓ DAS: Definition, Architecture, Working, Advantages, Disadvantages, Applications. ✓ NAS: Definition, Architecture, File-Level Storage, Protocols, Advantages, Disadvantages, Applications. ✓ SAN: Definition, Architecture, Block-Level Storage, Fibre Channel, iSCSI, Advantages, Disadvantages, Applications. ✓ DAS vs NAS vs SAN Comparison.	0.5 Day
8.	✓ Backup Types ✓ Backup Concept, Full Backup, Incremental Backup, Differential Backup, Backup Frequency, Backup Storage, Backup Restoration, Backup Window, Full vs Incremental vs Differential, Backup Strategy.	0.5 Day
9.	✓ Disaster Recovery ✓ Disaster Recovery (DR), Disaster Types, Disaster Recovery Plan, Business Continuity, RPO (Recovery Point Objective), RTO (Recovery Time Objective), Backup & Recovery, DR Site, Hot Site, Warm Site, Cold Site, RPO vs RTO.	0.5 Day
10.	✓ High Availability vs Fault Tolerance ✓ High Availability (HA), Fault Tolerance (FT), Availability, Downtime, Redundancy, Failover, Automatic Failover, Single Point of Failure, HA Architecture, FT Architecture, HA vs FT, Use Cases.	1 Day
11.	✓ Virtualization & Hypervisor ✓ Virtualization Concept, Server Virtualization, Virtual Machine (VM), Host Machine, Guest OS, Hypervisor, Type 1 Hypervisor (Bare-Metal), Type 2 Hypervisor (Hosted), CPU Virtualization, Memory Virtualization, Storage Virtualization, Network Virtualization,	1